

PROJECT DESIGN PHASE

Problem – Solution Fit

 **DATE**
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 **NAME**
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 **PROJECT**
**OptiCrop – Smart
Agricultural
Production
Optimization Engine**

 **MAX MARKS**
2 Marks

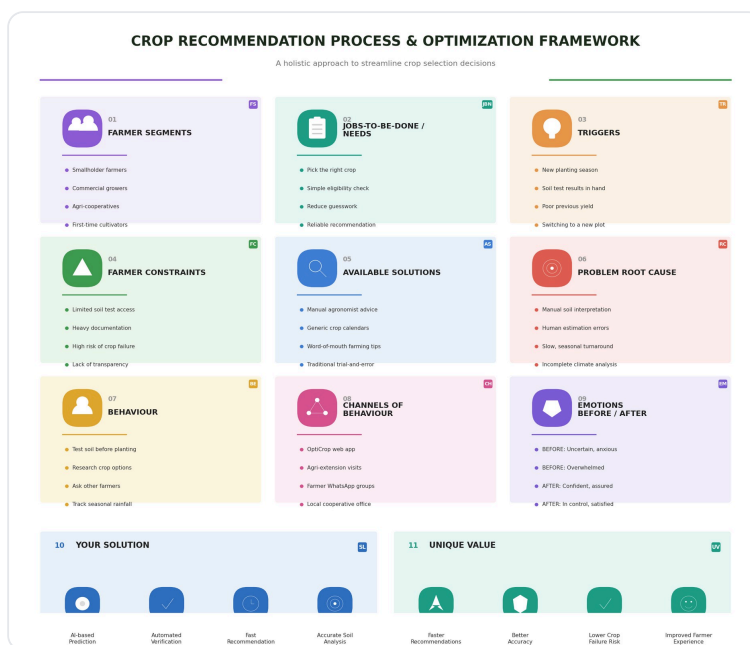
Project Solution Fit

The problem-solution fit means identifying the challenges faced by farmers and agricultural stakeholders in selecting suitable crops, and providing a machine-learning-based solution that accurately recommends the best crop for given soil and climate conditions, reducing guesswork and improving yield outcomes.

Main Purpose

- Reduces reliance on guesswork and improves crop selection accuracy.
- Improves farmer confidence with data-backed recommendations.
- Helps researchers and policymakers analyze crop-environment patterns.
- Enhances resource efficiency for water, fertilizer, and soil nutrients.
- Supports sustainable and evidence-based farming decisions.
- Recommends the crop best suited to a specific plot's nutrient and climate profile.
- Helps farmers plan seasonal crop choices in advance.

Figure 1 — Crop Recommendation Process & Optimization Framework



Crop Recommendation Process & Optimization Framework



Input

Soil nutrients (N, P, K) and climate readings (temperature, humidity, pH, rainfall).



Processing

Clustering and classification analysis on soil/climate patterns.



Output

Best-suited crop recommendation with a short suitability description.